LIBBY CASE REPORT

osteosarcoma-mets-dll3-h7r2

Multi-agent decision support — clinician track

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Executive summary

Case:osteosarcoma-mets-dll3-h7r2**Question:**Metastatic osteosarcoma with user-reported DLL3 RNA expression. What interventions could target DLL3 expression, gated on IHC confirmation? **Evidence base:**3 trials, 3 clinical-evidence rows, 4 preclinical rows. Two ranked rows: a workup gate plus one biomarker-conditional therapeutic option. Board agreement: full consensus on the workup; one persistent dissent (critic) on the conditional rank-2 trial.

What this report covers

The PI's synthesis frames the case around the DLL3 IHC test that gates whether tarlatamab via NCT06788938 is on the table. The case is scoped to drugs that target the user's stated targetable feature (DLL3); standard 2L+ care for the indication lies outside scope and is not enumerated.

Top-line findings

- DLL3 IHC SP347 is the load-bearing test. RNA expression alone does not establish membrane DLL3, and every DLL3-directed therapy in clinical use requires IHC. All five personas converged on this without dissent.
- The conditional rank-2 option is tarlatamab via NCT06788938 — a basket trial that includes osteosarcoma. The conservative's earlier toxicity veto lifts on biomarker confirmation.
- The critic's dissent on the trial persists: there are no published osteosarcoma data with tarlatamab, and cross-tumor translation from SCLC is unproven. That's the scientific question this enrollment would actually answer.
- **If IHC is negative:**rank 2 is foreclosed and the case has no within-scope recommendations. Libby's ranking is targetable-feature-scoped — the standard-of-care conversation for the indication is a separate one with the treating team and is not enumerated here.

Recommendation summary

Shared first step:DLL3 IHC (SP347) on tumor — gates whether rank 2 is reachable. All 5 personas endorse.

1. **Tarlatamab via NCT06788938**— *Conditional on DLL3 IHC positive — foreclosed if test is negative.*Strong preference fit (high-risk-high-reward + prefers trials); critic dissent on cross-tumor translatability persists. The only therapeutic option within scope of this case's targetable feature.

What this report does *not* cover

Standard 2L+ care for the indication is out of scope — Libby is a targetable-feature ranker, not a standard-of-care concierge. Dose adjustments, monitoring schedules, sequencing across lines, payer/access logistics, and institution-specific trial slot availability are also outside scope.

How to use this report

The ranked option carries the board's agreement state (endorsed, dissent, or veto by persona) and 1–3 anchor citations. The biomarker-conditional rank-2 option is flagged inline; the cross-cutting caveat at the top of the case page maps what's foreclosed if the test is negative. If IHC is negative, the next conversation with the treating team is the standard-of-care conversation for the indication — separate from Libby's targetable-feature ranking.

Libby is an experimental decision-support tool. The recommendations on this page have not been reviewed by a clinician treating this patient. Do not act on this report without consulting a qualified oncologist.

<!-- libby:downloads:begin -->

Downloads

- Clinician PDF report— ranked recommendations + evidence + sources
- Patient/caregiver PDF— plain-language summary
- Master manuscripts table (PDF)— every paper considered — n, effect, variance, toxicities
- Master manuscripts table (web)— same inventory in a sortable in-browser table
- Self-contained HTML— recommendations table that opens offline

<!-- libby:downloads:end -->

Research question

In metastatic osteosarcoma after first-line MAP, what interventions can target DLL3, gated on IHC confirmation?

Patient profile (scrubbed)

- **Primary site / histology:**bone — osteosarcoma
- **Stage:**IV (metastatic)
- **Performance status (assumed):**ECOG 1
- **Age band (assumed):**18-29 (typical osteosarcoma demographics; not user-supplied)
- **Sex:**unknown
- **Biomarkers:****DLL3 — RNA only (confirmation_status: rna_only); IHC SP347 status unknown.** Decision-relevant resolution: ≥1% (preferably ≥25%) by IHC for DLL3-directed clinical trials.
- **Prior therapy (assumed):**MAP frontline; response not provided.

Preferences

- **Efficacy/toxicity weight:**0.85 (strong efficacy lean)
- **Toxicity vetoes:**none stated
- **Modality constraints:**none stated
- **Free text:**"accepts high-risk high-reward options"
- **Trials preferred:**yes

Scope summary

Three trials, three clinical-evidence rows, four preclinical rows. Two ranked rows: a workup gate at rank 1 plus one biomarker-conditional therapeutic option at rank 2. The board reached full consensus (all five personas) on the workup. One persistent dissent (critic) sits on the conditional rank-2 trial. The case is scoped to drugs that act on the user's stated targetable feature (DLL3); standard 2L+ care for the indication is out of scope and is not enumerated.

Cross-cutting caveat (read first)

The DLL3 RNA expression in user input does not establish membrane DLL3 protein. The DLL3 IHC SP347 test (rank 1) gates whether tarlatamab via NCT06788938 (rank 2) is on the table at all. Every DLL3-directed therapy in current clinical use requires IHC protein-level confirmation. RNA expression is necessary but not sufficient.

- **Rank 2 (tarlatamab)**is conditional on DLL3 IHC ≥1% (preferably ≥25%). It is the only therapeutic option within scope of this case's targetable feature.
- **If IHC is negative:**rank 2 is foreclosed and this case has no within-scope recommendations. Libby's ranking is targetable-feature-scoped; standard 2L+ care for the indication is a separate conversation with the treating team and is not enumerated on this page.
- **Workup logistics:**SP347 IHC is non-toxic, runs on archival tissue (no fresh biopsy required), takes one to three weeks, and costs almost nothing relative to a treatment cycle. Confirm assay availability at the treating institution.

Intervention grouping

- **DLL3-directed BiTEs (biomarker-conditional):**tarlatamab via NCT06788938. Cross-tumor efficacy anchor: DeLLphi-301 (PMID 37861218) and DeLLphi-304 (PMID 40454646).

Top interventions

Rank 1. DLL3 IHC (SP347) on tumor — diagnostic gate

Non-toxic. Resolves whether the DLL3-directed pathway is open. Required for trial NCT06788938.

Evidence base

NCT06788938 enforces DLL3 IHC ≥25% (stage 1) or ≥1% (stage 2) for enrollment. The basket-trial design uses IHC for a concrete reason: DLL3 RNA expression does not predict surface protein density at the level the BiTE needs. Mechanistic anchor: every DLL3-directed therapy in clinical development gates on protein-level confirmation, not transcript.

Likelihood of desired effect

The test resolves whether rank 2 is reachable. Non-toxic and cheap regardless of result.

Toxicity profile

- None. Lab test on tissue.

Counter-productive mechanisms / dissent

Board endorsement was unanimous. No persona dissented or vetoed.

Practical considerations

Archival or fresh tumor works. Confirm SP347 antibody assay availability at the treating institution. One to three week turnaround. The IHC is the precondition for any DLL3-directed action regardless of which therapy the patient ultimately pursues.

Why this rank

The IHC is the precondition for rank 2. The board treated it as the gate, not as a therapy.

Per-trial detail

Test / trial	Efficacy context	Toxicity	Reference
DLL3 IHC SP347 (assay)	Gates enrollment in DLL3-directed therapy trials	None — diagnostic	NCT06788938

Rank 2. Tarlatamab via NCT06788938

Conditional on dll3_ihc:positive. Foreclosed if test is negative.

Bispecific T-cell engager with strong cross-tumor efficacy data in SCLC. The user's stated preferences fit this option when the trial is reachable. One persistent dissent on cross-tumor translation.

Evidence base

NCT06788938 (single-arm phase 2 basket, Simon two-stage, n=29 planned) is the trial enrolling osteosarcoma at biomarker-positive resolution. The mechanistic case rests on cross-tumor evidence: DeLLphi-301 (PMID 37861218 ; ORR 40%, n=220 SCLC) and the confirmatory DeLLphi-304 phase 3 (PMID 40454646; OS HR 0.60, 95% CI 0.47–0.77, median OS 13.6 vs 8.3 mo). No published osteosarcoma data with tarlatamab exist.

Likelihood of desired effect

Assuming positive IHC, mechanism-fit is concordant. Whether SCLC's clinical effect transfers to osteosarcoma is the open question this trial enrollment would answer. The user's preferences (efficacy lean 0.85, accepts high-risk-high-reward, prefers trials) fit this option. **A negative IHC forecloses this rec entirely.**

Toxicity profile

- CRS in ~50% (mostly grade 1–2; grade ≥3 ~1% in SCLC)
- ICANS-like neurologic events ~10%
- Inpatient cycle-1 step-up dosing required for CRS mitigation
- Step-up dosing: 1 mg D1, 10 mg D8/D15, then 10 mg q2w (28-day cycles)

User has no toxicity vetoes; CRS and inpatient cycle-1 chair time are not flagged.

Counter-productive mechanisms / dissent

The critic's dissent persists. No published osteosarcoma data with tarlatamab. Cross-tumor translation from SCLC is unproven, and the trial's basket design exists to test that premise. The conservative's earlier toxicity veto (issued under the IHC-unconfirmed scenario) lifts on biomarker confirmation; its own rationale specified the veto was contingent on IHC. Concensusite's qualified-on-guideline-fit position upgrades to endorsement when the trial-enrollment principle is the relevant frame.

Practical considerations

- Trial open at NCT06788938 (recruiting). Confirm slot availability at the treating site.
- Trial enrollment is NCCN cat-1 for relapsed osteosarcoma regardless of mechanism.
- Inpatient cycle-1 monitoring required.
- Off-guideline for osteosarcoma per indication; the trial provides the regulatory pathway.

Why this rank

The only DLL3-directed therapeutic option on the table, conditional on biomarker confirmation. Foreclosed entirely if IHC is negative.

Per-trial detail

Therapeutic agent	Efficacy	Toxicity	Reference
Tarlatamab — NCT06788938 DLL3-IHC-selected basket (osteosarcoma included), n=29 planned	ORR endpoint at 18 mo (primary); no osteosarcoma data yet	Per SCLC mechanism: CRS ~50%, ICANS ~10%, inpatient C1	NCT06788938
Tarlatamab — DeLLphi-301 SCLC (cross-tumor mechanism evidence), n=220	ORR 40% (95% CI 29–52); mPFS 4.9 mo	CRS ~50% (G3+ ~1%); ICANS ~10%; G3+ TRAEs 30%	PMID 37861218
Tarlatamab — DeLLphi-304 SCLC (cross-tumor confirmatory), n=509	OS HR 0.60 (95% CI 0.47–0.77), p<0.001; median OS 13.6 vs 8.3 mo	G3+ TRAEs 24% vs 53% chemo arm — favorable vs comparator	PMID 40454646

Classes examined but not ranked

- **DLL3-directed ADCs (rovalpituzumab tesirine / Rova-T):**mechanistically a DLL3-targeting modality, but not procurable. AbbVie withdrew Rova-T after the TAHOE phase-3 SCLC trial showed worse OS than topotecan (PMID 33002438). Listed for mechanism context. Not actionable.

Ranked prioritization

<div class="trial-table-wrap"> <div class="trial-scroll"> <table class="trial-table"> <thead><tr><th>Rank</th><th>Intervention</th><th>Likelihood of effect</th><th>Toxicity burden</th><th>Counter-productive MoA</th><th>Overall</th></tr></thead> <tbody> <tr> <td>1</td> <td>DLL3 IHC (SP347) on tumor — diagnostic gate
<small>endorse: risktaker conservative critic concensusite advocate</small></td> <td>Diagnostic certainty — resolves whether the DLL3-directed pathway is reachable; required by NCT06788938 (≥25% stage 1, ≥1% stage 2).</td> <td>Low (none — diagnostic test on archival tissue)</td> <td>N/A (diagnostic, not therapeutic)</td> <td>Non-toxic precondition that gates rank 2 entirely; run regardless of which therapy is ultimately chosen.</td> </tr> <tr> <td>2</td> <td>tarlatamab via NCT06788938 (UCCC-01 basket) (conditional on dll3_ihc

positive)
<small>endorse: risktaker advocate conservative concensusite</small>
<small>dissent: critic</small></td> <td>Cross-tumor extrapolation: SCLC OS HR 0.60 (DeLLphi-304); ORR ~40% (DeLLphi-301). Osteosarcoma efficacy is the open question NCT06788938 will answer.</td> <td>Moderate (CRS ~50% mostly G1-2; CRS G≥3 ~1%; ICANS-like ~10%; inpatient cycle-1 step-up dosing required)</td> <td>Moderate (On-mechanism CNS bystander T-cell activation drives ICANS; possible DLL3 antigen-loss escape on repeated dosing)</td> <td>The only DLL3-directed option when IHC is positive — preference-aligned but cross-tumor translation untested; foreclosed if IHC negative.</td> </tr> </tbody> </table> </div> </div>

!!! note "Reading the columns" **Toxicity burden**is patient-level G3+ AE severity (Low / Moderate / High) summarized from the trial publications. **Counter-productive MoA**is the mechanism-level risk that the intervention's own pathway could blunt the therapeutic goal — distinct from patient AEs. The board's endorse / dissent / veto state appears as pills under each intervention; full per-persona rationale lives on the board page.

Caveats

- The evidence base for the conditional trial is small. NCT06788938 plans n=29 with no published efficacy data yet. The mechanistic basis is the cross-tumor SCLC evidence (DeLLphi-301 / 304), which is robust in SCLC but unproven in any other tumor type.
- Biomarker dependency: rank 2's eligibility assumes a binary IHC result. Indeterminate or weak-positive (1–24%) edge cases are not explicitly addressed; in practice, NCT06788938's stage-2 ≥1% threshold may permit enrollment.
- What would change the ranking:
    - A positive DLL3 IHC plus a head-to-head osteosarcoma cohort within the basket trial reading out would move tarlatamab from "mechanism unproven cross-tumor" to "evidence-supported" and tighten its rank-2 confidence.
    - A user toxicity veto on CRS or inpatient cycle-1 monitoring would foreclose tarlatamab even with positive IHC.
    - Slot unavailability at NCT06788938 sites would close the within-scope therapeutic pathway. The patient's residual options would then be the treating team's standard-of-care conversation, which lies outside Libby's targetable-feature scope and is not enumerated here.

Sources

PubMed (PMID):

- 37861218— Ahn et al., DeLLphi-301, *NEJM*2023
- 40454646— Mountzios et al., DeLLphi-304, *NEJM*2025
- 35983951— DLL3 IHC reference (cited in workup row)

ClinicalTrials.gov (NCT):

- NCT06788938— DLL3-IHC-selected basket including osteosarcoma (tarlatamab)
- NCT05060016— DeLLphi-301 (cross-tumor mechanism)

- NCT05740566— DeLLphi-304 (cross-tumor confirmatory)

Transparency artifacts

- Trial table— 3 rows, all 25 columns
- Evidence list— 3 clinical-evidence rows + 4 preclinical rows
- Tumor-board transcript— 5 positions, 20 cross-critiques (filtered to in-scope drugs at render time)
- Recommendations table— full ranked detail with biomarker-conditional flag
- Plain-language summary— patient/caregiver track

Run log

Authored May 2026. Inputs: targetable feature ("DLL3 RNA expression"), clinical descriptor ("metastatic osteosarcoma"), preference summary ("accepts high-risk high-reward options"). Re-rendered when Libby's case scope tightened to drugs that act on the user's stated targetable feature; out-of-scope drugs (standard care for the indication that does not act on DLL3) no longer enter the dossier or appear on any case surface. The cross-cutting caveat carries the negative-result foreclosure mapping. The standard-of-care conversation for the indication is the treating team's, not Libby's. Humanizer pass applied May 2026.

Sources

PubMed (PMID):

- 35983951
- 37861218
- 40454646

ClinicalTrials.gov (NCT):

- NCT06788938